

NILAVANAN S A

Robotics and Cloud Engineer

@ sanilavanan2003@gmail.com 8903645897 Chennai, India <https://portfilo-nila.web.app/> in nilavanan

SUMMARY

Robotics and Cloud Engineer with 1+ years of hands-on experience in Backend Development and Business Analysis. Skilled in Python with API Integrations , UiPath for IoT and RPA workflows. Learning to build scalable cloud-native systems with CI/CD, Docker, Kubernetes, and real-time monitoring. Passionate about combining robotics, AI, and cloud to create smart automation solutions for Industry 4.0.

EXPERIENCE

Backend Developer (Contract)

Tunelink Networks AI - HYBRID Aug 2025 - Present

Tech: Python, Django, SQL, Git

- **Scalable Systems:** Built high-availability backends for enterprise-level operations.
- **Optimization:** Improved database efficiency and ensured 100% data integrity.

Backend Developer (Intern)

Mittai Healthcare Pvt Ltd - ONSITE Jan 2025 - Jul 2025

Tech: Node.js, Grafana, Twilio, FastAPI

- **Automation:** Developed AI-voice systems to streamline business communications.
- **Insights:** Created real-time performance dashboards for management decision-making.

PROJECTS

Smart Manufacturing Automation using UiPath + GCP

UiPath, GCP (Pub/Sub, BigQuery, Vertex AI, Looker Studio), Python, IoT

Aug 2025 - Present

- Engineered an **Industry 4.0 system** integrating **IoT and RPA**; built real-time data pipelines (**Pub/Sub** → **BigQuery**) and trained a **Vertex AI model** for predictive maintenance.
- Automated industrial workflows and reporting via **UiPath bots** and **Looker Studio**, deploying a scalable cloud-native solution on **Google Cloud Platform**.

Autonomous Document Intelligence with AWS + RPA

AWS (Lambda, Textract, Bedrock, S3, DynamoDB), UiPath, Python, SNS

Sept 2025 - Present

- Architected a **serverless AI pipeline** using **AWS Lambda, Textract, and Bedrock** to process unstructured documents, reducing manual entry by **90%** and increasing throughput by **40%**.
- Integrated **UiPath bots** for ERP synchronization and automated exception handling via **SNS**, leveraging **DynamoDB** for secure, scalable data storage.

EDUCATION

MBA (Financial Management)[DDE]

ANNAMALAI UNIVERSITY

Aug 2025 - May 2027 (In Progress)

B.E Computer Science (Data Science)

ANNAMALAI UNIVERSITY

Nov 2021 - May 2025 OGPA: 8.05

SKILLS

Python

UiPath

MSPowerAutomate

Arduino

Raspberry Pi

Linux

Sensors & Actuators

OpenCV

Tensorflow

Google Cloud

AWS

API Integrations

PromptEngineering

GenAI

VibeCoding

PostgresSQL

Git

GitLab

Github

RAG

ACHIEVEMENTS

- Served as a dedicated student representative for 4 consecutive years . 2021-2024
- Achieved a prestigious 5-star rating on Hackerank for both Java and SQL .

CERTIFICATION

- Diploma in Robotics - Alison [Ongoing]
- **Deep Learning (IIT Ropar) -Swayam**
- **Advanced Distributed Systems - Swayam**
- **Predictive Analysis - Canva**
- **SQL (Advanced) - HackerRank**